7. A particle of mass m is fixed to one end of a light spring of force constant k and un-stretched length l. It is rotated with an angular velocity ω in horizontal circle. What will be the length increase in the spring?

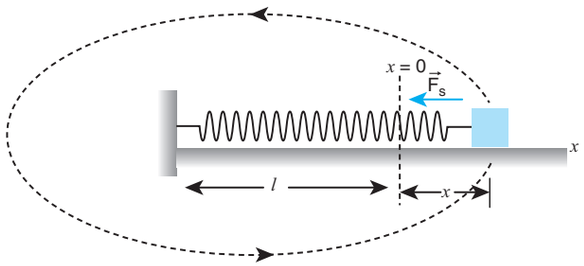
Solution:

Mass of particle = m

Force constant = k

Un-stretched length = l

Angular velocity = ω



Let 'x' be the increase in the length of the spring.

The new length = (l+x) = r

When the spring is rotated in a horizontal circle,

Spring force = centripetal force.

$$kx = m\omega^2(l+x)$$

$$x = \frac{m\omega^2 l}{k - m\omega^2}$$

8. A gun fires 8 bullets per second into a target X. If the mass of each bullet is 3 g and its speed 600 ms^{-1}, then calculate the power delivered by the bullets.

Solution:

Power = work done per second = total kinetic energy of 8 bullets per second

$$P = 8 \times (\text{kinetic energy of each bullet per second})$$

$$= 8 \times \frac{1}{2} \times \left(3 \times 10^{-3}\right) \times \left(600\right)^2$$

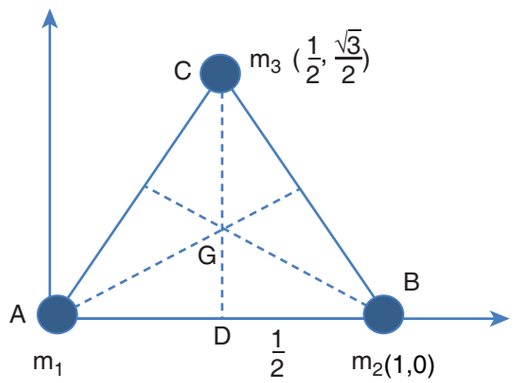
P = 4320 W

P = 4.320 kW

SOLVED EXAMPLE UNIT-5

1. Three particles of masses m_1 = 1 kg, m_2 = 2 kg and m_3 = 3 kg are placed at the corners of an equilateral triangle of side 1m as shown in Figure. Find the position of centre of mass.

Solution:



The centre of mass of an equilateral triangle lies at its geometrical centre G.

The positions of the mass m_1, m_2 and m_3 are at positions A, B and C as shown in the Figure.

From the given position of the masses, the coordinates of the masses m_1 and m_2 are easily marked as (0,0) and (1,0) respectively.

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